

Number, units and estimation - Set 1 - Solutions

Specification M1.1, M1.2, M2.1-M2.14.

Answer key

Question	Answer	Topic
1	D	Number, units and estimation
2	C	Number, units and estimation
3	D	Number, units and estimation
4	F	Number, units and estimation
5	F	Number, units and estimation
6	D	Number, units and estimation
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8	E	Number, units and estimation
9	D	Number, units and estimation
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12	F	Number, units and estimation

Worked solutions

Question 1 NUM-01 · Number, units and estimation · spec M2.3

Find the highest common factor of $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 7$.

Answer: D

Take the lower power of each common prime: $2^2 \times 3^2 = 36$.

Full method - how and why

HCF from prime factorisations: take each **common** prime to its **lower** power.

- 1 List the primes in each: $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 7$.

why: Both are already factorised, which is the form the rule needs.

- 2 Take the common primes only: 2 and 3 appear in both; 5 and 7 do not.

why: A prime missing from one number cannot divide both.

- 3 Take the lower power of each: $2^{\min(3,2)} = 2^2$ and $3^{\min(2,3)} = 3^2$.

why: The HCF must divide both, so it cannot use more copies of a prime than the smaller number has.

- 4 Multiply: $4 \times 9 = 36$.

why: Check: 36 divides 360 and 756 ✓.

Where it goes wrong

- Taking the higher powers (that gives the LCM contribution, not the HCF).
- Including 5 or 7, which gives a number dividing only one of them.

Takeaway HCF: common primes, lowest powers. LCM: all primes, highest powers.

Question 2 NUM-03 · Number, units and estimation · spec M2.9

Express the recurring decimal $0.\dot{2}7$ as a fraction in its lowest terms.

Answer: C

$$x = 0.2727\dots, 100x = 27.2727\dots, \text{ so } 99x = 27 \text{ and } x = \frac{27}{99} = \frac{3}{11}.$$

Full method - how and why

The standard algebraic trick for recurring decimals: multiply by a power of 10 that shifts one full period, then subtract.

- 1 Let $x = 0.272727\dots$. The repeating block has **two** digits, so multiply by $10^2 = 100$: $100x = 27.272727\dots$

why: Shifting by exactly one period makes the decimal tails identical, so they cancel.

- 2 Subtract: $100x - x = 27.2727\dots - 0.2727\dots = 27$, so $99x = 27$.

why: The infinite tails cancel exactly - this is the whole method.

- 3 Solve: $x = \frac{27}{99}$.

why: The denominator is always $10^k - 1$, i.e. a string of nines matching the period length.

- 4 Simplify by 9: $\frac{3}{11}$.

why: The question demands lowest terms, and $\frac{27}{99}$ is offered as a trap.

Where it goes wrong

- Choosing $\frac{27}{99}$ - correct in value but not in lowest terms.
- Multiplying by 10 instead of 100, which does not align the tails.

Takeaway Multiply by $10^{(\text{period length})}$, subtract, then simplify fully.

Question 3 NUM-04 · Number, units and estimation · spec M2.13

A length is given as 4.6 cm correct to 1 decimal place, and a width as 3 cm correct to the nearest cm. Find the upper bound of the area.

Answer: D

Upper bounds 4.65 and 3.5: $4.65 \times 3.5 = 16.275$.

Full method - how and why

Bounds: each measurement is an interval. For a maximum product, take the top of each interval.

- Length 4.6 to 1 d.p. means $4.55 \leq \ell < 4.65$, so the upper bound is 4.65.
why: Rounding to 1 d.p. means the true value is within 0.05 either side.
- Width 3 to the nearest cm means $2.5 \leq w < 3.5$, so the upper bound is 3.5.
why: Nearest whole number means within 0.5 either side - a different half-width from the length.
- Both factors are positive, so the area is largest when both are largest: 4.65×3.5 .
why: Products of positive numbers increase with each factor.
- Compute: $4.65 \times 3.5 = 16.275 \text{ cm}^2$.
why: $4.65 \times 3 = 13.95$, plus $4.65 \times 0.5 = 2.325$.

Where it goes wrong

- Using $4.65 \times 3.0 = 13.95$ by forgetting the width also has a bound.
- Using half-widths of 0.05 for both, giving 3.05 for the width.

Takeaway Each measurement has its own accuracy: 1 d.p. gives ± 0.05 , nearest whole gives ± 0.5 .

Question 4 NUM-05 · Number, units and estimation · spec M2.8

Evaluate $\frac{(3 \times 10^4) \times (8 \times 10^{-2})}{6 \times 10^{-3}}$, giving your answer in standard form.

Answer: F

$$\frac{24 \times 10^2}{6 \times 10^{-3}} = 4 \times 10^5.$$

Full method - how and why

Standard-form arithmetic: handle the numbers and the powers of ten separately, then renormalise.

- Multiply the numerator: $(3 \times 8) \times 10^{4+(-2)} = 24 \times 10^2$.
why: Multiply the leading numbers, add the indices.
- Divide by 6×10^{-3} : $\frac{24}{6} \times 10^{2-(-3)} = 4 \times 10^5$.
why: Divide the numbers, subtract the indices - and subtracting -3 adds 3.
- Check the form: $1 \leq 4 < 10$ ✓, so no renormalising is needed.
why: Standard form requires the leading number to be between 1 and 10.

Where it goes wrong

- Getting 10^{-1} by adding instead of subtracting the -3 .
- Leaving 24×10^2 , which is not in standard form.

Takeaway Numbers multiply/divide; indices add/subtract. Then check $1 \leq a < 10$.

Question 5 NUM-06 · Number, units and estimation · spec M2.5

A code consists of two letters from A, B, C, D, E followed by two digits from 1 to 9. Letters may be repeated but the two digits must be different. How many codes are possible?

Answer: F

$$5 \times 5 \times 9 \times 8 = 1800.$$

Full method - how and why

The product rule for counting, with one part allowing repeats and the other not.

- 1 Letters: 5 choices for the first and, since repeats are allowed, 5 again for the second: 5×5 .

why: 'May be repeated' means the second choice is unaffected by the first.

- 2 Digits: 9 choices for the first, but only 8 for the second because they must differ: 9×8 .

why: 'Must be different' removes exactly one option from the second slot.

- 3 Multiply all the stages: $5 \times 5 \times 9 \times 8 = 1800$.

why: The product rule: independent stages multiply.

Where it goes wrong

- Using $9 \times 9 = 81$ for the digits, giving 2025 - an offered option.
- Using 5×4 for the letters, giving 1440.

Takeaway Multiply the number of choices at each stage, reducing by one only where repetition is forbidden.

Question 6 NUM-07 · Number, units and estimation · spec M1.2

A car travels at 90 km/h. What is this speed in metres per second?

Answer: D

$$90\,000 \text{ m in } 3600 \text{ s: } 90000/3600 = 25 \text{ m/s.}$$

Full method - how and why

A compound-unit conversion: convert distance and time separately.

- 1 $90 \text{ km} = 90 \times 1000 = 90\,000 \text{ m}$.

why: Convert the distance unit first.

- 2 $1 \text{ hour} = 60 \times 60 = 3600 \text{ s}$.

why: Convert the time unit.

- 3 Divide: $\frac{90\,000}{3600} = 25 \text{ m/s}$.

why: Equivalently, divide km/h by 3.6 - worth memorising.

Where it goes wrong

- Multiplying by 3.6 instead of dividing, giving 324 - an offered option.
- Converting only one of the two units.

Takeaway $\text{km/h} \div 3.6 = \text{m/s}$; convert numerator and denominator units separately.

Question 7 NUM-08 · Number, units and estimation · spec M2.14

Estimate the value of $\frac{4.92 \times \sqrt{99.3}}{0.198}$ by rounding each number to 1 significant figure.

Answer: B

$$\frac{5 \times 10}{0.2} = 250.$$

Full method - how and why

Estimation by rounding to one significant figure, choosing roundings that make the arithmetic exact.

- 1 Round each: $4.92 \approx 5$, $99.3 \approx 100$, $0.198 \approx 0.2$.

why: One significant figure; 99.3 rounds to 100, which is also a perfect square - convenient.

- 2 Take the root: $\sqrt{100} = 10$.

why: Choosing 100 rather than 99 is what makes the root exact.

- 3 Compute: $\frac{5 \times 10}{0.2} = \frac{50}{0.2} = 250$.

why: Dividing by 0.2 is multiplying by 5.

Where it goes wrong

- Dividing by 0.2 as if it were 2, giving 25.
- Forgetting the square root and using 100 directly, giving 2500.

Takeaway Round to 1 s.f., but choose the rounding that makes roots and divisions exact; dividing by 0.2 multiplies by 5.

Question 8 NUM-10 · Number, units and estimation · spec M2.13

A number x is truncated (not rounded) to 2 decimal places and the result is 3.14. Which inequality describes the possible values of x ?

Answer: E

Truncation removes the later digits, so x is at least 3.14 and less than 3.15.

Full method - how and why

Truncation, not rounding. Truncating throws away everything after the cut, so the value can only be **above** the stated one.

- 1 Truncating to 2 d.p. means simply deleting the third and later decimals.

why: No rounding up ever happens, which makes the interval one-sided.

- 2 The smallest such x is 3.14 itself (with nothing after it).

why: 3.14 truncates to 3.14, so equality is allowed at the bottom.

- 3 The largest is anything below 3.15: at 3.15 the truncation would read 3.15.

why: Strict inequality at the top.

- 4 So $3.14 \leq x < 3.15$.

why: Contrast with rounding, which would give $3.135 \leq x < 3.145$ - an offered distractor.

Where it goes wrong

- Using the rounding interval $3.135 \leq x < 3.145$.
- Making the lower end strict, excluding 3.14 itself.

Takeaway Truncation: $[a, a + 10^{-n})$. Rounding: $[a - \frac{1}{2}10^{-n}, a + \frac{1}{2}10^{-n})$.

Question 9 NUM-11 · Number, units and estimation · spec M2.2

Evaluate $\left(2\frac{1}{3} - 1\frac{3}{4}\right) \div \frac{7}{6}$.

Answer: D

$$\frac{7}{3} - \frac{7}{4} = \frac{28-21}{12} = \frac{7}{12}; \frac{7}{12} \times \frac{6}{7} = \frac{1}{2}.$$

Full method - how and why

Mixed numbers and division of fractions. Convert to improper fractions, subtract, then multiply by the reciprocal.

- 1 Convert: $2\frac{1}{3} = \frac{7}{3}$ and $1\frac{3}{4} = \frac{7}{4}$.
why: Whole part times denominator plus numerator.
- 2 Subtract with a common denominator 12: $\frac{28}{12} - \frac{21}{12} = \frac{7}{12}$.
why: $\frac{7}{3} = \frac{28}{12}$ and $\frac{7}{4} = \frac{21}{12}$.
- 3 Divide by $\frac{7}{6}$ means multiply by $\frac{6}{7}$: $\frac{7}{12} \times \frac{6}{7}$.
why: Dividing by a fraction is multiplying by its reciprocal.
- 4 Cancel and multiply: the 7s cancel and $\frac{6}{12} = \frac{1}{2}$.
why: Cancel before multiplying to keep numbers small.

Where it goes wrong

- Multiplying by $\frac{7}{6}$ instead of its reciprocal, giving $\frac{49}{72}$ - an offered option.
- Subtracting the whole parts and fractions separately and mishandling the borrow.

Takeaway Improper fractions, common denominator, then multiply by the reciprocal - and cancel before multiplying.

Question 10 NUM-13 · Number, units and estimation · spec M2.7

Which of the following is equal to $\frac{8^4 \times 2^{-5}}{4^3}$?

Answer: B

$$\frac{2^{12} \cdot 2^{-5}}{2^6} = 2^1 = 2.$$

Full method - how and why

Everything is a power of 2. Convert and use the index laws.

- 1 Convert: $8 = 2^3$ so $8^4 = 2^{12}$; $4 = 2^2$ so $4^3 = 2^6$.
why: A common base turns the whole expression into index arithmetic.
- 2 Numerator: $2^{12} \times 2^{-5} = 2^7$.
why: Multiplying adds indices.
- 3 Divide: $2^{7-6} = 2^1 = 2$.
why: Dividing subtracts indices.

Where it goes wrong

- Using $8^4 = 2^7$ (adding instead of multiplying the indices).
- Sign slip on the -5 , giving $2^{11-6} = 32$.

Takeaway Rewrite every number as a power of the same base, then just add and subtract indices.

Question 11 NUM-14 · Number, units and estimation · spec M2.12Simplify $\sqrt{48} \times \sqrt{27} \div \sqrt{12}$.**Answer: F**

$$\sqrt{48} = 4\sqrt{3}, \sqrt{27} = 3\sqrt{3}, \sqrt{12} = 2\sqrt{3}. \text{ So } \frac{4\sqrt{3} \times 3\sqrt{3}}{2\sqrt{3}} = \frac{36}{2\sqrt{3}} = \frac{18}{\sqrt{3}} = 6\sqrt{3}.$$

Full method - how and whySimplify each surd first; they all reduce to multiples of $\sqrt{3}$, so the expression collapses.

- 1 Simplify: $\sqrt{48} = \sqrt{16 \times 3} = 4\sqrt{3}$, $\sqrt{27} = 3\sqrt{3}$, $\sqrt{12} = 2\sqrt{3}$.

why: Take out the largest square factor each time.

- 2 Multiply the first two: $4\sqrt{3} \times 3\sqrt{3} = 12 \times 3 = 36$.

why: $\sqrt{3} \times \sqrt{3} = 3$ - the surd disappears.

- 3 Divide: $\frac{36}{2\sqrt{3}} = \frac{18}{\sqrt{3}}$.

why: Only one surd remains, in the denominator.

- 4 Rationalise: $\frac{18}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{18\sqrt{3}}{3} = 6\sqrt{3}$.

why: Multiply top and bottom by $\sqrt{3}$.**Where it goes wrong**

- Answering 18 by forgetting to rationalise (or by cancelling the $\sqrt{3}$ incorrectly).
- Combining under one root: $\sqrt{48 \times 27 \div 12} = \sqrt{108}$ - correct but then needing simplification to $6\sqrt{3}$ anyway.

Takeaway Simplify each surd, do the arithmetic, then rationalise any surd left in a denominator.**Question 12** NUM-15 · Number, units and estimation · spec M2.5

How many three-digit positive integers have digits that are all different and all odd?

Answer: FOdd digits 1, 3, 5, 7, 9: $5 \times 4 \times 3 = 60$.**Full method - how and why**

Counting with two restrictions: digits from a fixed set, and all different.

- 1 List the allowed digits: 1, 3, 5, 7, 9 - five odd digits.

why: 'All odd' restricts the pool; note 0 is even so no leading-zero problem arises.

- 2 First digit: 5 choices. Second: 4 (one used). Third: 3.

why: 'All different' removes one option at each stage.

- 3 Multiply: $5 \times 4 \times 3 = 60$.

why: The product rule.

Where it goes wrong

- Using $5^3 = 125$ by ignoring 'all different' - an offered option.
- Worrying about leading zeros, which cannot occur here.

Takeaway Count position by position, reducing the pool by one each time when repetition is banned.